

CSOC IG NLP WEEK 2

Replication Study: Scaling Laws for Neural Language Models

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June 30, 2026

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Abstract

This project replicates key experiments from Scaling Laws for Neural Language Models (Kaplan et al. 2020). A decoder only transformer was built entirely from scratch in pytorch, without using high level attention modules. The implementation includes a custom causal multi-head attention, feed forward neural networks and pre-layer normalization, learnable positional embeddings, and weights between the token embedding and the language model head. Then the scaling experiments were conducted to analyze the relationship between validation loss and both model size and dataset size. By fitting power-law equations to our experimental data using log-log linear regression, we estimated the scaling exponents $\alpha_N \approx 0.064$ and $\alpha_D \approx 0.14$, and compared with the results reported in the original paper.

1 Mathematical Formalism

1.1 Causal Multi-Head Attention (MHA)

X is the input tensor of shape (B, T, d_model), the query (Q), key(K), and value(V) are defined as

$$Q = XW_q \quad K = XW_k \quad V = XW_v \quad (1)$$

The attention is computed as

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M \right) V \quad (2)$$

Here M is lower triangular causal mask preventing the attention to future tokens.

1.2 Position-wise Feed-Forward Network (FFN)

This is a two-layer MLP mapping, each sequence independently passes through,

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2 \quad (3)$$

where W_1 is of shape (d_model, d_ff) and W_2 is of shape (d_ff, d_model), with $d_{ff} = 4 \times d_{\text{model}}$, which is taken as said.

1.3 Pre-Layer Normalization (Pre-LN)

The pre-ln is defined as,

$$x^{(k+1)} = x^{(k)} + \text{SubLayer}(\text{LN}(x^{(k)})) \quad (4)$$

1.4 Positional Embeddings

Learnable positional embedding are added to token embeddings before entering the Transformer layer. This was chosen over the sinusoidal ones because this was easier to implement and performed well in the chosen dataset.

1.5 Weight tying

Output language model head shares its weights with the token embedding matrix to reduce parameter count.

2 Replication Methodology

2.1 Dataset and Preprocessing

Dataset chosen was Tiny Shakespeare.

- tokenization was done Character level.
- The dataset was split as 90% training
- 10% for validation.

Dataset was processed simple enough, it had 65 unique characters, and around 1 million characters for training set and 100k for testing.

2.2 Optimization Configuration

The chose optimizer was AdamW

- learning rate of 10^{-3}
- Batch size of 32
- Total training steps of 3000

The AdamW was chosen over standard adam for proper penalization of large weights. Batch size of 32 made the traning process fast and with efficient vRAM consumption.

2.3 Sweep Design

2.3.1 Parameter Scaling sweep

Three decoder only transformers were trained.

Table 1: Model Configurations for Parameter Sweep

Model	Layers (L)	Hidden Size (d_{model})	Heads (H)	Context Length	Parameters (N)
Tiny	2	64	2	128	1.0×10^5
Small	4	128	4	256	7.9×10^5
Medium	6	256	8	256	4.7×10^6

2.3.2 Data Scaling sweep

The Small model configuration from parameter scalling was trained here using

- 10%
- 25%
- 50%
- 100%

of the training data. Each training started with a freshly initialized model with Small configuration above, and trained for 3000 optimization steps.

3 Empirical Results and Scaling Curves

3.1 Parameter Scaling Sweep

The best validation loss is given below for all the three models. We can see the Validation loss decreases as the Parameters(N) increases, smoothly. They are fit according to the power-law equation in the data in log-log space, it is done using polyfit from numpy.

$$\log(L(N)) = \text{constant} - \alpha_N \log(N)$$

Table 2: Model Configurations for Parameter Sweep

Model	Parameters (N)	Best Validation Loss L(N)
Tiny	1.0×10^5	2.10
Small	7.9×10^5	1.87
Medium	4.7×10^6	1.65

The plot of the equation line is given below

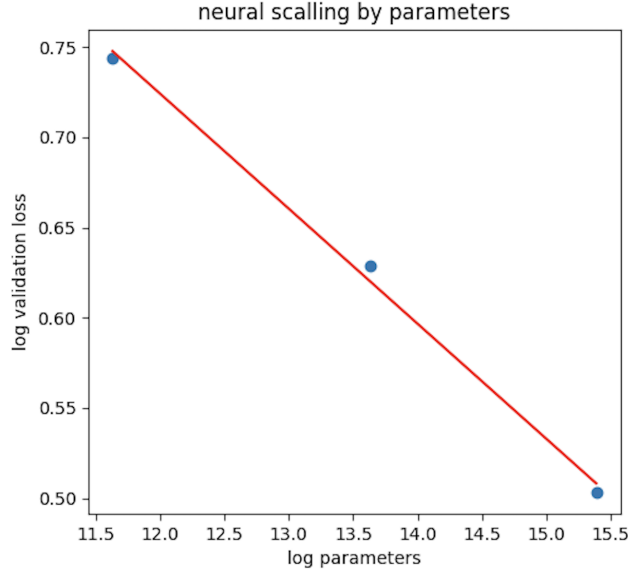


Figure 1: The fit regression line on Parameter scaling sweep results

From the equation the exponent α_N is the negative of the slope which was obtained as approx. 0.063.

3.2 Data Scaling Sweep

The best validation loss is given below for all the 4 scaled datasets. We can see the Validation

Table 3: Model Configurations for Parameter Sweep

Data Subset	Approx. Length of Data	Best Validation Loss L(D)
10%	100k	2.6
25%	250k	2.36
50%	500k	2.13
100%	1 million	1.88

loss decreases smoothly as the Length of the training data is increasing. They are fit according to the power-law equation below in log-log space, done similarly as did above.

$$\log(L(D)) = \text{constant} - \alpha_D \log(D)$$

The plot of the equation is given below.

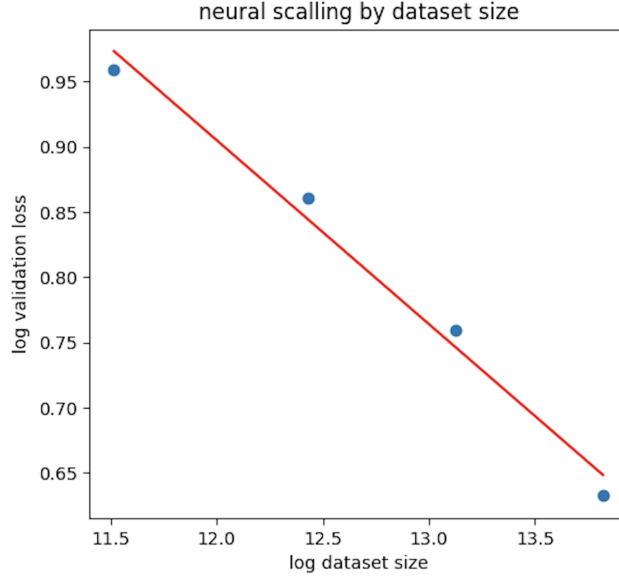


Figure 2: The fit regression line on Data scaling sweep results

From the equation the exponent α_D is the negative of the slope, which was obtained to be approx. 0.14.

4 Theoretical Discussion

In this replication study, the parameter scaling exponent and the dataset size scaling exponent were calculated to be approx. $\alpha_N \approx 0.063$ and $\alpha_D \approx 0.14$. Use these values we can determine the critical scaling ratio γ .

$$\gamma = \frac{\alpha_N}{\alpha_D} = \frac{0.063}{0.14} \approx 0.45 \quad (5)$$

In the foundational paper Scaling Laws for Neural Language Model (Kaplan et al., 2020), reported the exponentsof $\alpha_N \approx 0.076$ and $\alpha_D \approx 0.095$, yielding a scaling ratio of $\gamma \approx 0.80$. Our results aligns remarkably well with their findings proving that the predictable power-law nature of parameter scaling holds true even for a small character-level model.

But, however our empirical scaling ratio γ differs notably from the original paper. A ratio of $\gamma < 1.0$ mathematically indicates that the architecture is currently operating in a severely data-constrained regime. The high data exponent of $\alpha_D \approx 0.14$ relative to the α_N indicates that the marginal increase in dataset size yield substantially faster validation loss improvements than equivalent increases in model capacity.

We assume that this behavior is tied fundamentally to the choice of dataset. The Tiny Shakespeare contains roughly 1 million characters, when it was trained with 10% of the data, the model performed worse and was overfit because of the small text size, but when the dataset size was scaled up it learned much faster, resulting in the higher α_D than the original research paper, were it was trained on a very large dataset (WebText2), so the smaller dataset size scaled models did not perform very worse there.

References

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